

ESSLLI ‘Semantic properties and combinatorial restrictions of attitude predicates’

Day 4

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1. INTRODUCTION

■ In this course, we have seen that attitude predicates differ in their combinatorial properties:

- clause types:
 - (1) Responsives: ✓declarative ✓interrogative
Cleo knows {that, whether} Petra is happy.
 - (2) Anti-rogatives: ✓declarative ✗interrogative
Cleo believes {that, *whether} Petra is happy.
 - (3) Rogatives: ✗declarative ✓interrogative
Cleo wonders {*that, whether} Petra is happy.
- clause *subtypes*, e.g. declarative clause subtypes in Romance:
 - (4) Cleo thinks that Petra {is.Ind, *is.Subj} happy. ✓indicative ✗subjunctive
 - (5) Cleo wants / wishes that Petra {*is.Ind, is.Subj} happy. ✗indicative ✓subjunctive

■ Recent literature strives to derive clause-type restrictions from the semantic characterization of the attitude predicate:

- veridical → responsive (Egré 2008)
- change of state → responsive (White & Rawlins 2019)
- non-veridical + preferential → anti-rogative (Uegaki & Sudo 2019)
- neg-raising → anti-rogative (Zuber 1982, Egré 2008, Mayr 2019, Theiler et al. 2019)

■ Similarly, there is an effort to derive mood-based clause-*subtype* restrictions from the semantics of the attitude predicate. Here we will consider two approaches:

- **Approach A:** Comparison-based approach (Giorgi & Pianesi 1997)
- (6) If an attitude predicate involves a non-null Ordering Source, it selects Subjunctive.
- **Approach B:** Context Set-based (Schlenker 2005)
- (7) If an attitude predicate evaluates its complement clause only for the worlds within the so-called local context set (roughly, its Modal Base), it selects Indicative.

■ Today:

We will dive into mood clause subtypes in Spanish and compare approaches A and B.

2. MOOD IN SPANISH: A (PARTIAL) OVERVIEW

- Like other Romance languages, (Iberian) Spanish exhibits a contrast between indicative and subjunctive mood in several constructions:

- Mood selection by attitude verbs INTENSIONAL MOOD
- Mood alternation under negated indicative-selecting verbs POLARITY MOOD
- Mood alternation in adjunct clauses: RELATIVE clauses, CONDITIONAL clauses, a.o.

■ INTENSIONAL mood:

Some attitude verbs select indicative in their complement clause and some select subjunctive, leading to a grammaticality contrast (Quer 1998, Villalta 2008):

- (8) Sofia sabe / piensa / sueña que Ana planea.IND / *planee.SUBJ un picnic.
Sofia **knows** / **thinks** / **dreams** that Ana plans.IND / *plans.SUBJ a picnic
'Sofia knows / thinks / dreams that Ana plans a picnic.'
- (9) Sofia quiere / ordena que Ana *planea.IND / planee.SUBJ un picnic.
Sofia **wants** / **commands** that Ana *plans.IND / plans.SUBJ a picnic
'Sofia wants / commands Ana to plan a picnic.'

■ POLARITY mood:

Indicative-selecting verbs, when embedded under negation, allow for mood alternation. As we will see, this alternation leads to a clear interpretive contrast in cases like (11)-(12) but not in other cases (Montero & Romero 2023):

- (10) Sofia no sabe / piensa que Ana planea.IND / planee.SUBJ un picnic.
Sofia not **knows** / **thinks** that Ana plans.IND / plans.SUBJ a picnic
'Sofia doesn't know / thinks that Ana plans a picnic.'
- (11) Sofia doesn't **know** [that Ana plans.IND a picnic]. ↗ Ana plans a picnic.
- (12) Sofia doesn't **know** [that Ana plans.SUBJ a picnic]. ↘ Ana plans a picnic.

■ Mood alternation in RELATIVE clauses:

Under subjunctive-selecting verbs, Relative Clauses alternate between indicative and subjunctive, which has been argued to lead to a de re vs. de dicto distinction:

- (13) Sofia **quiere** que Ana compre un libro [que tiene.IND las tapas rojas].
Sofia **wants** that Ana buys.SUBJ a book [that has.IND the covers red]
'Sofia wants Ana to buy a specific book that has a red cover.' ⇔ De re reading
- (14) Sofia **quiere** que Ana compre un libro [que tenga.SUBJ las tapas rojas].
Sofia **wants** that Ana buys.SUBJ a book [that has.SUBJ the covers red]
'Sofia wants Ana to buy some book or other –any would do– that has a red cover.'
⇔ De dicto reading

■ Mood alternation in CONDITIONAL clauses:

If-clauses alternate between indicative and subjunctive, leading to different modal interpretations:

- (15) Si Ana compró.**IND** un libro ayer, Sofia se alegró.
 If Ana bought.IND a book yesterday, Sofia SE rejoiced
 ‘If Ana bought a book yesterday, Sofia was happy.’ ⇔ Epistemic
- (16) Si Ana hubiese.**SUBJ** comprado un libro ayer, Sofia se habría alegrado.
 If Ana had.SUBJ bought a book yesterday, S. SE would.have rejoiced
 ‘If Ana had bought a book yesterday, Sofia would have been happy.’ ⇔ Counterfactual

■ As a desideratum, the field aims at a theory of mood that is:

- (i) valid for a given construction across different languages
 ↳ E.g. Portner & Rubinstein (2020) on ‘hope’ vs ‘want’ across Romance
- (ii) valid across constructions within a given language (in this case, Spanish)
 ↳ Main concern of this talk

■ Point of departure:

After a long tradition (Farkas 1992, Giannakidou 1994, Quer 1998, Villalta 2008, among many others), two main approaches to INTENSIONAL mood emerge:

Approach A: Comparison-based (Giorgi & Pianesi 1997, Portner & Rubinstein 2020)

Approach B: Context Set-based (von Stechow 1997, Schlenker 2005, a.o.)

■ Primary goal for today:

To combine analytical ingredients from these two approaches to allow for a uniform treatment across the grammar of Spanish:

- for Intensional Mood selected by attitude verbs
- for mood alternation in Relative Clauses
- for mood alternation in Conditional Clauses

■ Secondary goal of the today (**time permitting!**):

To advance our understanding of Polarity Mood

Joint work:
 Montero & Romero (2023)

■ Roadmap for the present talk

- §2 Combining ingredients from different analyses
- §3 Intensional Mood
- §4 Motivation of the choice of ingredients
- §5 Polarity Subjunctive (**time permitting!**)
- §6 Outlook

2. COMBINING INGREDIENTS FROM DIFFERENT ANALYSES

- Back to our two main approaches to intensional mood:

Approach A: Comparison-based (Giorgi & Pianesi 1997, Portner & Rubinstein 2020)

Approach B: Context Set-based (von Stechow 1997, Schlenker 2005, a.o.)

- Our proposal will combine ingredients from both:

- ① Presuppositional content: $\Leftrightarrow$ **Approach A**
Indicative is used when the domain of quantification involves one modal background;
subjunctive is used when the domain of quantification involves two modal backgrounds.
- ② Locus of quantification: $\Leftrightarrow$ **Approach B**
Quantification over worlds is not introduced by mood itself, but by the attitude verb
(or by some Force operator).
- ③ Existence of a semantically unmarked form: $\Leftrightarrow$ **Approach B**
Indicative mood is presuppositionally constrained (thus, marked form), whereas
subjunctive mood is not presuppositionally constrained (thus, unmarked form).

2.1. Introducing ingredient ①

- Conversational backgrounds (Kratzer 1991):

A conversational background is a set of propositions (cf. accessibility relation).

Conversational backgrounds come in different flavors: (17).

- (17) Conversational backgrounds:

$\text{Dox}_x(w) =$ the set of propositions that x *believes* in w to be true.
 $\text{Bou}_x(w) =$ the set of propositions that x *desires* in w to be(come) true.
 $\text{Deo}(w) =$ the set of propositions that conform to *what the law provides* in w .
 $\text{Circ}(w) =$ the set of prop. that describe the *actual facts/circumstances* in w .
 $\text{Epi}_x(w) =$ the set of propositions that x *knows* in w to be true.
 ...

- Hintikka-style semantics for belief verbs (Hintikka 1969):

Attitude verbs introduce quantification over the domain of worlds arising from the relevant conversational background, used as MODAL BASE.

- (18) $\llbracket x \text{ believes } p \rrbracket$
 $= \lambda w_0. \forall w \in \cap \text{Dox}_x(w_0) [p(w)]$
 $=$ we are in a world w_0 such that, for each of x 's belief worlds w in w_0 , p is true in that w .

- (19) Bea **believes** that John teaches Semantics.
 $\lambda w_0. \forall w [w \in \cap \text{Dox}_{\text{bea}}(w_0) \rightarrow \text{John teaches semantics in } w]$

- Comparative semantics for desire verbs (Heim 1992, Villalta 2008, Romero 2015, a.o.):
Besides a MODAL BASE, a second conversational background is used as ORDERING SOURCE to establish a desirability ranking ($>$) among worlds.

- (20) Intuitive idea:
 x wants p means
"among x 's belief worlds, the most desirable ones are the ones in which p is true".
- (21) For any $w', w'' \in W$:
 $w' <_{\text{Bou}_x(w_0)} w''$ iff w' is more desirable according to $\text{Bou}_x(w_0)$ than w'' .
- (22) $\text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0))$
 $= \{w': w' \in \cap \text{Dox}_x(w_0) \wedge \neg \exists v [v \in \cap \text{Dox}_x(w_0) \wedge v <_{\text{Bou}_x(w_0)} w']\}$
 $=$ the set of x 's belief worlds that are best according to the ranking by $\text{Bou}_x(w_0)$
- (23) $\llbracket x \text{ wants } p \rrbracket$
 $= \lambda w_0. \forall w \in \text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0)) [p(w)]$
- (24) Bea **wants** John to teach Semantics.
 $\lambda w_0. \forall w \in \text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0)) [\text{John teaches sem in } w]$

- Ingredient ① of the proposal: SIMPLEX vs. COMPLEX content of s

(slightly adapted from Portner & Rubinstein 2020)

- Event-relativity of modal backgrounds (Kratzer 2006, Hacquard 2006, a.o.):¹
- (25) Peter believes that p .
 $\lambda w_0. \exists s [s <_{w_0} \wedge \text{believing}(s) \wedge \text{Experiencer}(s, \text{peter}) \wedge \forall w \in \text{Dom}(\text{content}(s)) [p(w)]]$
- (26) Peter wants that p .
 $\lambda w_0. \exists s [s <_{w_0} \wedge \text{wanting}(s) \wedge \text{Experiencer}(s, \text{peter}) \wedge \forall w \in \text{Dom}(\text{content}(s)) [p(w)]]$
- (27) a. If s is a believing situation, $\text{content}(s)$ is a single background Dox .
b. If s is a wanting situation, $\text{content}(s)$ is a pair of backgrounds $\langle \text{Dox}^+, \text{Bou} \rangle$.
- Simplex vs complex content of an attitude situation s :
- (28) For any attitude situation s and any modal backgrounds f and h :
a. $\text{Content}(s)$ is **SIMPLEX** iff $\text{content}(s)$ consists of a single background f .
b. $\text{Content}(s)$ is **COMPLEX** iff $\text{content}(s)$ consists of a pair of backgrounds $\langle f, h \rangle$.
- Building a domain Dom out of $\text{content}(s)$:
- (29) For any modal backgrounds f and h , any world w and any attitude holder x :
a. $\text{Dom}(f_x(w)) = \cap f_x(w)$
b. $\text{Dom}(\langle f_x(w), h_x(w) \rangle) = \text{BEST}_w(\cap f_x(w), h_x(w))$

¹ I leave open whether $\text{Dom}(\text{content}(s))$ and the $\forall w$ -quantification in (25)-(26) is introduced by the attitude verb itself, as in traditional approaches, or by the Force operator in the embedded clause (Kratzer 2006, Hacquard (2006). For the sake of simplicity in this presentation, I will use the former possibility.

2.2. Introducing ingredient ②

■ Pronominal approach to tense (Partee 1973, Kratzer 1998):

- A personal pronoun refers to an individual provided that its gender(/number) precondition is satisfied: (30)
- Similarly, tense morphology has been argued to refer to an temporal interval provided that its temporal precondition is met. That is, tense morphology does not introduce quantification over times, but acts merely as a temporal “pronoun” that can be bound by a temporal operator higher up in the tree: (31)

(30) $\llbracket she_2 \rrbracket^g$ is defined only if $g(2)$ is female;
if defined, $\llbracket she_2 \rrbracket^g = g(2)$

(31) $\llbracket Past_2^{pro1} \rrbracket^g$ is defined only if $g(2)$ temporally precedes $g(1)$;
if defined, $\llbracket Past_2^{pro1} \rrbracket^g = g(2)$

■ Ingredient ② of the proposal: Pronominal approach to mood (Schlenker 2005)

Similarly, mood morphology refers to a world provided that its “modal” precondition is satisfied. That is, mood morphology does not introduce quantification over worlds, but merely acts as a world “pronoun” that can be bound higher up.

(32) a. $\llbracket IND_2^{pro1} \rrbracket^g$ is defined only if ...???... ; [To be revised]
if defined, $\llbracket Ind_2^{pro1} \rrbracket^g = g(2)$
b. $\llbracket SUBJ_2^{pro1} \rrbracket^g$ is defined only if ...???... ;
if defined, $\llbracket Subj_2^{pro1} \rrbracket^g = g(2)$

■ Combining ingredients ① and ② of the proposal:

Take $g(1)$ to refer to the attitude situation introduced by the matrix attitude verb (i.e., the believing situation s , the wanting situation s , etc.). Then:

(33) a. $\llbracket IND_2^{pro1} \rrbracket^g$ is defined only if content($g(1)$) is SIMPLEX and
 $g(2) \in \text{Dom}(\text{content}(g(1)))$;
if defined, $\llbracket Ind_2^{pro1} \rrbracket^g = g(2)$
b. $\llbracket SUBJ_2^{pro1} \rrbracket^g$ is defined only if content($g(1)$) is COMPLEX and
 $g(2) \in \text{Dom}(\text{content}(g(1)))$;
if defined, $\llbracket Subj_2^{pro1} \rrbracket^g = g(2)$ [To be revised]

2.3. Introducing ingredient ③

- Some expressions come in pairs including a presuppositional and a non-presuppositional version:

- (34) a. $\llbracket \text{the} \rrbracket^g = \lambda P_{\langle e, t \rangle}: \text{there is exactly one } x \text{ such that } P(x). \lambda Q_{\langle e, t \rangle}. \exists x [P(x) \wedge Q(x)]$
 b. $\llbracket a \rrbracket^g = \lambda P_{\langle e, t \rangle}: \text{there is more than one } x \text{ such that } P(x). \lambda Q_{\langle e, t \rangle}. \exists x [P(x) \wedge Q(x)]$

- (35) [Context: S(peaker) meets Felix at a conference, who identifies himself as a PhD student of Maribel. S does not know exactly how many PhD students Maribel has. S says:]
 S: I met a PhD student of Maribel at the conference.

- Maximize presupposition:

- (36) Maximize presupposition: Make your contribution presuppose as much as possible.
 (Heim 1991)

- (37) [S is a knowledgeable person on earth:]
 a. S: The sun is shining.
 b. S: # A sun is shining.

- Ingredient ③ of the proposal: Subjunctive as unmarked, non-presuppositional
 (Schlenker 2005)

- (38) a. $\llbracket \text{IND}_2^{\text{prol}} \rrbracket^g$ is defined only if ??? ; [To be revised]
 if defined, $\llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2)$
 b. $\llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g$ ~~is defined only if ??? ;~~
~~if defined, $\llbracket \text{Subj}_2^{\text{prol}} \rrbracket^g = g(2)$~~

- Combining ingredients ①, ② and ③ of the proposal:
 Take $g(1)$ to refer to the immediate domain of quantification. Then:

- (39) a. $\llbracket \text{IND}_2^{\text{prol}} \rrbracket^g$ is defined only if $\text{content}(g(1))$ is SIMPLEX and $g(2) \in \text{Dom}(\text{content}(g(1)))$;
 if defined, $\llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2)$
 b. $\llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g$ $g(2)$ [Final version]

UPSHOT 1

- (40) $\llbracket \text{Ana plans.IND}_2^{\text{prol}} \text{ a picnic} \rrbracket^g =$
 $\lambda w: \text{content}(g(1)) \text{ is SIMPLEX and } w \in \text{Dom}(\text{content}(g(1))). \text{ Ana plans a picnic in } w$
 (41) $\llbracket \text{Ana plans.SUBJ}_2^{\text{prol}} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{ Ana plans a picnic in } w$

3. INTENSIONAL SUBJUNCTIVE

3.1. Selected mood in complement clauses

■ Verbs selecting an INDICATIVE complement clause in Spanish (Villalta 2008):

- Doxastic (/epistemic) verbs: e.g. *saber* 'know', *pensar* 'think'
- Verbs of communication: e.g. *decir* 'say'
- Verbs of certainty: e.g. *estar seguro de* 'be certain that'
- Verbs of commitment: e.g. *prometer* 'to promise'.
- Verbs of fiction: e.g. *soñar* 'to dream'.
- Verbs of mental judgement: e.g. *adivinar* 'to guess'
- Perception verbs: e.g. *ver* 'to see'

- (42) Sofia sabe / piensa / sueña que Ana planea.IND / *planee.SUBJ un picnic.
Sofia **knows** / **thinks** / **dreams** that Ana plans.IND / *plans.SUBJ a picnic
'Sofia knows / thinks / dreams that Ana plans a picnic.'
(=8)

■ Verbs selecting a SUBJUNCTIVE complement clause in Spanish (Villalta 2008):

- Desire verbs: e.g. *querer* 'to want', *temer* 'to fear', *esperar* 'hope'
- Factive-emotive verbs: e.g. *alegrarse de* 'to be glad that', ...
- Modal predicates: e.g. *ser posible* 'to be possible'
- Verbs of doubt: e.g. *dudar* 'to doubt'.
- Directive verbs: e.g. *ordenar* 'to order'
- Causative verbs: e.g. *hacer* 'to make (somebody do sth.)'

- (43) Sofia quiere / ordena que Ana *planea.IND / planee.SUBJ un picnic.
Sofia **wants** / **commands** that Ana *plans.IND / plans.SUBJ a picnic
'Sofia wants / commands Ana to plan a picnic.'
(=9)

■ Empirical generalization:

All subjunctive-selecting verbs involve comparisons via an ordering source *h*, whereas all indicative-selecting verbs do not invoke an ordering source (Giorgi & Pianesi 1997, Villalta 2008, Portner & Rubinstein 2020):

- (44) Indicative-selecting verbs: (Portner & Rubinstein 2020: 355)

Verb	Modal Base <i>f</i>
'believe'	Dox
'know'	Epi
'say'	Rpg (reported Common Ground)
'dream'	Drm (content of the dream)

- (45) Subjunctive-selecting verbs: (Portner & Rubinstein 2020: 355)

Verb	Modal Base <i>f</i>	Ordering Source <i>h</i>
'want'	Dox ⁺	Bou
'order'	Rpg	Deo
'regret'	Dox	Bou ⁻
'be probable'	Circ	Ste

■ Applying the proposal to a sample **indicative-selecting verb**: ‘think’

- Since ‘think’ introduces a simplex content of s –namely, $\text{Dox}_x(w_0)$ –, it suffices to have the partial proposition (48) defined for the corresponding domain. $\Leftrightarrow$ **✓IND**
- Although the total proposition (49) could in principle also compose with the attitude verb, Maximize Presupposition demands that presuppositionally heavier (48) be used. $\Leftrightarrow$ ***SUBJ**

(46) Sofia thinks that Ana plans **IND** / ***plans.SUBJ** a picnic.

(47) $\llbracket \text{think} \rrbracket = \lambda p. \lambda x. \lambda w_0. \exists s [s < w_0 \wedge \text{thinking}(s) \wedge \text{Exp}(s, x) \wedge \forall w \in \cap \text{Dox}_x(w_0) [p(w)]]$

(48) $\llbracket \text{Ana plans.IND}_2^{\text{Dox}_x(w_0)} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{Dox}_x(w_0) \text{ is SIMPLEX and } w \in \cap \text{Dox}_x(w_0). \text{ Ana plans a picnic in } w$

(49) $\llbracket \text{Ana plans.SUBJ}_2^{\text{Dox}_x(w_0)} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{ Ana plans a picnic in } w$

■ Applying the proposal to a sample **subjunctive-selecting verb**: ‘want’

- Since ‘want’ introduces a complex content of s –namely, $\langle \text{Dox}_x(w_0), \text{Bou}_x(w_0) \rangle$ –, the presupposition of the indicative proposition (52) is not satisfied. This means that (52) is not defined for any possible world, leading to a presupposition failure when composed with the verb. $\Leftrightarrow$ ***IND**
- In contrast, the total subjunctive proposition (53) felicitously composes with the verb. $\Leftrightarrow$ **SUBJ**

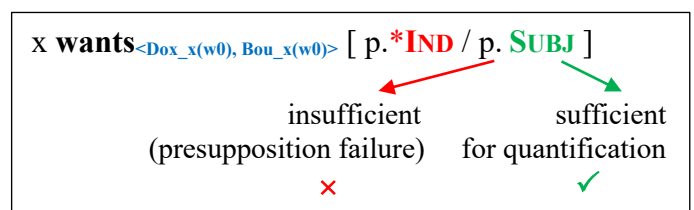
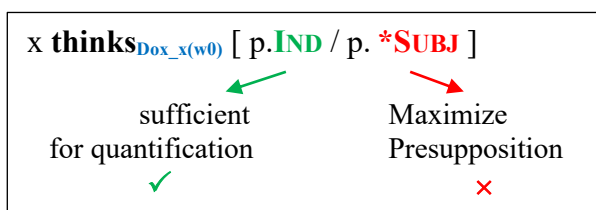
(50) Sofia wants that Ana ***plans.IND** / plans **SUBJ** a picnic.

(51) $\llbracket \text{want} \rrbracket = \lambda p. \lambda x. \lambda w_0. \exists s [s < w_0 \wedge \text{wanting}(s) \wedge \text{Exp}(s, x) \wedge \forall w \in \text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0)) [p(w)]]$

(52) $\llbracket \text{Ana plans.IND}_2^{\langle \text{Dox}_x(w_0), \text{Bou}_x(w_0) \rangle} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \langle \text{Dox}_x(w_0), \text{Bou}_x(w_0) \rangle \text{ is SIMPLEX and } w \in \text{BEST}_{w_0}(\cap \text{Dox}_x(w_0), \text{Bou}_x(w_0)). \text{ Ana plans a picnic in } w$

(53) $\llbracket \text{Ana plans.SUBJ}_2^{\langle \text{Dox}_x(w_0), \text{Bou}_x(w_0) \rangle} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{ Ana plans a picnic in } w$

UPSHOT 2



3.2. Extension to matrix clauses

- Though matrix clauses usually appear in the indicative, they can also appear in the subjunctive, leading to different types of sentential force (Portner 2018, a.o.):

- (54) Ana planea.**IND** un picnic. Statement
 Ana plans.**IND** a picnic
 ‘Ana plans a picnic.’
- (55) a. Planea.SUBJ un picnic. Order
 Plan.**SUBJ** a picnic
 ‘Plan a picnic.’
 b. Que la Fuerza te acompañe.SUBJ. Wish
 That the force te.ACC accompanies.**SUBJ**
 ‘May the Force be with you.’
 c. ¡Que Ana esté.SUBJ despierta tan tarde! Exclamation (surprise)
 That Ana is.**SUBJ** awake so late
 ‘That Ana be awake so late!’

- Operator analysis of sentential force (Bierwisch 1980, Krifka 2001, 2014; see Portner 2018 for alternatives):

- (56) ASSERT [p.IND] Statement
- (57) a. COMMAND [p.SUBJ] Order
 b. DESIRE [p.SUBJ] Wish
 c. SURPRISE [p.SUBJ] Exclamation (surprise)

- Applying the proposal to a matrix sentence headed by **ASSERT**:

- The partial, indicative proposition in (60) is defined for all the worlds that need to be quantified over and thus suffices to carry out the desire quantification. ⇔ **✓IND**
- By Maximize Presupposition, the presuppositionally heavier IND must be used. ⇔ ***SUBJ**

- (58) ASSERT [Ana plans.**IND** / *plans.**SUBJ** a picnic].

- (59) $\llbracket \text{ASSERT} \rrbracket = \lambda p. \lambda x. \lambda w_0. \exists s [s < w_0 \wedge \text{asserting}(s) \wedge \text{Agent}(s, x) \wedge \forall w \in \cap \text{Rpg}_x(w_0) [p(w)]]$

- (60) $\llbracket \text{Ana plans.IND}_2^{\cap \text{Dox}(w_0)} \text{ a picnic} \rrbracket^g =$
 $\lambda w: \text{Rpg}_x(w_0) \text{ is SIMPLEX and } w \in \cap \text{Rpg}_x(w_0). \text{ Ana plans a picnic in } w$

- (61) $\llbracket \text{Ana plans.SUBJ}_2^{\cap \text{Dox}(w_0)} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{ Ana plans a picnic in } w$

■ Applying the proposal to a matrix sentence headed by **COMMAND**:

- Since **COMMAND** introduces a complex content of *s*, the presupposition of the indicative proposition (64) is not satisfied and leads to a presupposition failure.
⇒ ***IND**
- In contrast, the total subjunctive proposition (65) felicitously composes with the verb.
⇒ **SUBJ**

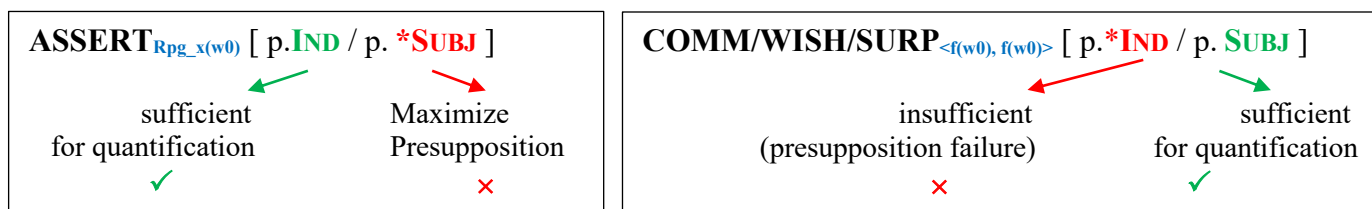
(62) **COMMAND** [Ana ***plans.IND** / plans.**SUBJ** a picnic].

(63) $\llbracket \text{COMMAND} \rrbracket = \lambda p. \lambda x. \lambda w_0. \exists s [s < w_0 \wedge \text{commanding}(s) \wedge \text{Agent}(s, x) \wedge \forall w \in \text{BEST}_{w_0}(\cap \text{Rpg}_x(w_0), \text{Deo}_x(w_0)) [p(w)]]$

(64) $\llbracket \text{Ana plans.IND}_2^{\text{BEST}_{w_0}(\cap \text{Rpg}_x(w_0), \text{Deo}_x(w_0))} \text{ a picnic} \rrbracket^g =$
 $\lambda w: \langle \text{Rgp}_x(w_0), \text{Deo}_x(w_0) \rangle \text{ is SIMPLEX and } w \in \text{BEST}_{w_0}(\cap \text{Rpg}_x(w_0), \text{Deo}_x(w_0)). \text{ Ana plans a picnic in } w$

(65) $\llbracket \text{Ana plans.SUBJ}_2^{\text{BEST}_{w_0}(\cap \text{Rpg}_x(w_0), \text{Deo}_x(w_0))} \text{ a picnic} \rrbracket^g =$
 $\lambda w. \text{ Ana plans a picnic in } w$

UPSHOT 3



4. MOTIVATION FOR THE CHOICE OF INGREDIENTS

4.1. Choice of Ingredient ①

- For the first ingredient, we sided with the **approach A**, where IND signals that the referent of the world pronoun is anchored to a domain with a simplex content of s:

$$\begin{aligned}
 (66) \quad a. \llbracket \text{IND}_2^{\text{prol}} \rrbracket^g & \text{ is defined only if } \text{content}(g(1)) \text{ is SIMPLEX and } \\
 & \text{ } g(2) \in \text{Dom}(\text{content}(g(1))); \\
 & \text{ if defined, } \llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2) \\
 b. \llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g & g(2) \quad (=39)
 \end{aligned}$$

- In contrast, **approach B** anchors the referent of the world pronoun to the local Context Set CS (in the sense of Stalnaker 1984):

- For matrix clauses, $\text{CS} = \cap \{p: p \text{ is a proposition in the Common Ground}\}$
- For complement clauses
 $\text{CS} = \text{the great intersection of some representational modal background}$
 $=_{\text{e.g.}} \cap \text{Dox}_x(w_0)$
 $=_{\text{e.g.}} \cap \text{Drm}_x(w_0)$
 $\dots$

$$\begin{aligned}
 (67) \quad a. \llbracket \text{IND}_2^{\text{prol}} \rrbracket^g & \text{ is defined only if } g(1) \text{ is a local CS and } g(2) \in g(1); \\
 & \text{ if defined, } \llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2) \\
 b. \llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g & g(2)
 \end{aligned}$$

- The Context Set-based ingredient ① derives the correct results for:

- Indicative-selecting verbs
- Subjunctive-selecting verbs whose lexical semantics operates on a superset of the local CS, e.g. *be happy* in (68) and *want* in (70):

$$\begin{aligned}
 (68) \quad \llbracket x \text{ is happy that } p \rrbracket & \\
 = \lambda w_0: \forall w \in \cap \text{Dox}_x(w_0) [p(w)]. & \\
 \quad \forall w \in \cap \text{Dox}_x(w_0) [w <_{\text{Bou}_x(w_0)} \text{Sim}_w(\text{rev}_p(\text{Dox}_x(w_0)) \dashv p)] & \\
 = \text{presupposes that } x \text{ believes } p \text{ and} & \\
 \text{asserts that, for each belief world } w \text{ of } x: w \text{ is more desirable than its } & \text{not-}p \\
 \text{counterpart world} &
 \end{aligned}$$

⇒ This counterpart world is outside of the local context set $\cap \text{Dox}_x(w_0)$.
Hence, *alegrarse de* ‘be happy that’ is correctly predicted to select subjunctive.

(69) I live in Paris and I want to leave in Paris.

(70) $\llbracket x \text{ wants that } p \rrbracket$ (Heim 1992)
 $= \lambda w_0. \forall w \in \cap \text{Dox}_x(w_0) [\text{Sim}_w(\text{Dox}_x^+(w_0)+p) <_{\text{Bou}_x(w_0)} \text{Sim}_w(\text{Dox}_x^+(w_0)+\neg p)]$
 $=$ for each belief world w of x :
 the most similar world to w where **certain beliefs of x** and p are true
 is more desirable than
 the most similar world to w where those same beliefs but not p are true

$\Leftrightarrow$ To avoid triviality for statements like (69), only some beliefs of x are used (indicated as **$\text{Dox}_x^+(w_0)$**), not all of them (i.e., not $\text{Dox}_x(w_0)$) (Heim 1992, Rubinstein 2017).
Hence, *querer* ‘want’ is correctly predicted to select subjunctive (Romero 2012).

■ However, the Context Set-based ingredient ① derives the incorrect results for:

- Subjunctive-selecting verbs whose lexical semantics operates only on the local CS, e.g. *hope*, *be possible/probable/necessary*:

(71) # I live in Paris and I hope to leave in Paris. (Portner & Rubinstein 2020)

(72) $\llbracket x \text{ hopes that } p \rrbracket$ (Heim 1992)
 $= \lambda w_0. \forall w \in \cap \text{Dox}_x(w_0) [\text{Sim}_w(\text{Dox}_x(w_0)+p) <_{\text{Bou}_x(w_0)} \text{Sim}_w(\text{Dox}_x(w_0)+\neg p)]$
 $\Leftrightarrow$ No operation in this lexical entry involves going beyond the local Context Set $\cap \text{Dox}_x(w_0)$.
Hence, *esperar* ‘hope’ would incorrectly be predicted to select indicative in Spanish under **approach B**.

- **Approach A** would derive the two readings above as well, but it would also derive the unattested reading in (80) for the subjunctive version (76):³

- (80) Unattested reading:
 $\lambda w_0. \forall w \in \cap Rpg_x(w_0)$
 $\exists x [\text{book}_w(x) \wedge \forall w' \in \text{BEST}_w(\cap \text{Dox}_{\text{sofia}}(w), \text{Bou}_{\text{sofia}}(w)) [\text{has-red-cover}_w(x)] \wedge$
 $\forall w' \in \text{BEST}_w(\cap \text{Dox}_{\text{sofia}}(w), \text{Bou}_{\text{sofia}}(w)) [\text{Ana buys}_{sw'} x]]$
 ‘There is a book x such that: Sofia wants x to have a red cover and Sofia wants Ana to buy x.’

4.3. Choice of Ingredient ③

- For the third ingredient, we sided with **approach B** and made SUBJ the unmarked, presuppositionally vacuous form:

- (81) a. $\llbracket \text{IND}_2^{\text{prol}} \rrbracket^g$ is defined only if $\text{content}(g(1))$ is **SIMPLEX** and $g(2) \in \text{Dom}(\text{content}(g(1)))$;
 if defined, $\llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2)$
 b. $\llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g$ $g(2)$ (=39)

- In contrast, **approach A** makes SUBJ as marked / contentful as IND:

- (82) a. $\llbracket \text{IND}_2^{\text{prol}} \rrbracket^g = \lambda p. \lambda s: \text{Content}(s)$ consists of a single modal background f.
 $\forall w \in \cap f [p(w)]$
 b. $\llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g = \lambda p. \lambda s: \text{Content}(s)$ consists of two modal backgrounds g and h.
 $\forall w \in \text{BEST}_s(f, h) [p(w)]$

- Suppletive form:
 Morphologically, subjunctive is known to be the suppletive form when there is a gap in the paradigm (Schlenker 2005, a.o.):

- (83) Llama a Lucía.
 Call.IMPER.2sg A Lucia
 ‘Call Lucia.’

- (84) Llame a Lucía.
 Call.SUBJ.3sg A Lucia
 ‘(Sir,) Call Lucia.’

⇒ SUBJ as the “elsewhere” form, as in **approach B**.

³ See Alonso-Ovalle, Menéndez-Benito and Rubinstein (2022) on purpose-like Relative Clauses where another source of quantification is needed.

■ Mood in conditional clauses (Lewis 1973, von Stechow 1997):

- Mood alternation in conditionals leads to an interpretive contrast parallel to the two types of domains under attitudes:⁴

(85) Si Juan fue a la fiesta ayer, (pro) la disfrutó. EPISTEMIC
 If Juan went. **IND** to the party yesterday, (he) it enjoyed
 ‘If Juan went to the party yesterday, he enjoyed it.’

(86) $\lambda w_0. \forall w \in \cap \text{Epi}(w_0)$ [Juan went_w to party $\rightarrow$ John enjoyed_w it]

COUNTERFACTUAL
 (87) Si Juan hubiese ido a la fiesta ayer, la habría disfrutado.
 If Juan had. **SUBJ** gone to the party yesterday, it would. have enjoyed
 ‘If Juan went to the party yesterday, he enjoyed it.’

(88) $\lambda w_0. \forall w \in \text{BEST}(\cap \text{Cir}(w_0), \text{Rea}(w_0))$ [Juan went_w to party $\rightarrow$ John enjoyed_w it]⁵

- Interestingly, in certain pragmatic contexts, the interpretive effect of subjunctive is cancellable: Despite being in the subjunctive, the antecedent proposition in (89) is not understood as contrary-to-fact.

(89) If Jones had taken arsenic, he would be having right now the symptoms that he is indeed having. (... Hence, he probably took arsenic.) (Anderson 1951)

(90) Si Juan hubiese tomado arsénico, tendría ahora mismo los síntomas
 If Juan had. **SUBJ** taken arsenic, would. have now right the symptoms
 que está teniendo.
 that is having

$\Leftrightarrow$ The lack of counterfactuality in (90) can be derived if SUBJ is presuppositionally vacuous, as in **approach B**, and if the effects of Maximize Presupposition can be cancelled in certain pragmatic contexts.⁶

⁴ ‘Fake tense’ is necessary but not sufficient to build a counterfactual conditional in Spanish; subjunctive mood is needed as well (Romero 2017).

⁵ The semantics of counterfactual conditionals is rendered in a very simplified form in (88).

⁶ See Leahy (2011) for a derivation of counterfactuality as a (cancellable) anti-presupposition.

5.1. Litetature review

■ Recall our initial data on Polarity Mood:

(91) Sofia no sabe / piensa que Ana planea.IND / planea.SUBJ un picnic. (=10)
 Sofia not **knows** / **thinks** that Ana plans.IND / plans.SUBJ a picnic
 'Sofia doesn't know / thinks that Ana plans a picnic.'

(92) Sofia doesn't **know** [that Ana plans.IND a picnic]. $\rightsquigarrow$ Ana plans a picnic. (=11)

(93) Sofia doesn't **know** [that Ana plans.SUBJ a picnic]. $\nrightarrow$ Ana plans a picnic. (=12)

■ The literature (Rivero 1971, Quer 1998, Ridruejo 1999, Borgonovo 2003, Laca 2010, a.o.) mentions several verbs as giving rise to this interpretive difference:

- factive verbs: *saber* 'know', *recordar* 'remember', *darse cuenta* 'realize', ..., and
- non-factive predicates: *decir* 'say', *creer* 'believe', *comunicar* 'communicate', ...

■ The theoretical explanations given in the literature in principle apply to both verb classes:

- Quer (1998): In a tripartite quantificational structure Qu-Restrictor-NuclearScope (Partee 1991), indicative maps the complement p to the restrictor (hence presupposing its truth) whereas subjunctive fails to do so.
- Borgonovo (2003): indicative marks the matrix predicate as focus of negation, subjunctive signals that the focus of negation is the embedded clause.
- Buchczyk (2022) (for *no decir* 'not say'): Indicative signals that the complement p needs to be accommodated into the Common Ground (and thus that Speaker believes p); subjunctive gives no such signal.

■ However, there is no systematic investigation in the literature and some of the data points are debatable.

⇒ Need of more nuanced empirical exploration

5.2. Experimental study⁷

■ Design

- 3 x 2 Factorial Design: [verb class] x [mood]

Verb class	Mood		
	IND	SUBJ	
V1			V1 = 5 cognitive factive verbs (<i>remember, notice, know, find out, see</i>)
V2			V2 = 5 non-factive/non-fiction verbs (<i>say, tell, think, believe, suspect</i>)
V3			V3 = 5 fiction verbs (<i>dream, fantasize, invent, fake and make believe</i>)

- Structure of items: *Carmen didn't V1/V2/V3 that Mary arrived.IND/SUBJ on time.*

Did Mary arrive on time?

1 2 3 4 5

No ☐ ☐ ☐ ☐ ☐ Yes

- Materials: 30 critical items, 10 controls and 20 fillers counterbalanced across subjects using a Latin-square design.
- Participants: 50 native speakers of Peninsular Spanish recruited via Prolific.
- Platform: PennController platform (Zehr and Schwarz 2018)

■ Results:

- Mean scores for Speaker Commitment to embedded *p* (see Appendix for each verb):



Fig. 1

⁷ See Montero and Romero (2023) for more details on the study.

- Effects after running a linear mixed-effects regression model:
 - Two main effects: mood and verb class
 - Significant interaction between the two

	Chisq	Df	Pr(>Chisq)	
(Intercept)	547.14	1	< 2.2e-16 ***	
Mood	57.15	1	4.039e-14 ***	Main effect
VerbType	10.83	2	0.004449 **	Main effect
Mood:VerbType	20.12	2	4.276e-05 ***	Interaction

Table 1

- Post hoc Test for to measure significance of interaction:

Factive: class V1

contrast	p.value
ind - subj	<.0001

Non-factive/Non-fiction: class V2

contrast	p.value
ind - subj	0.0589

Fiction: class V3

contrast	p.value
ind - subj	0.0240

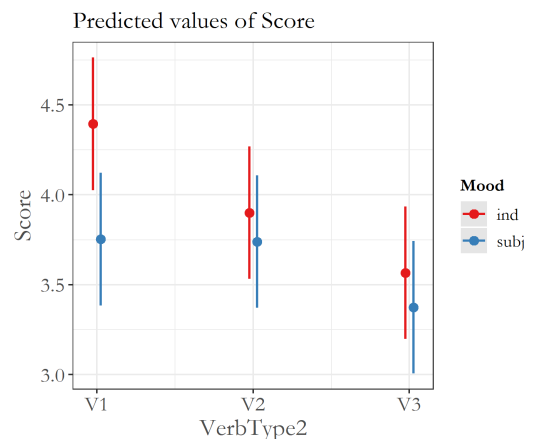


Fig. 2

UPSHOT 4

- (94)
- For factive/**V1 verbs**, there is a **significant effect of mood choice**:
The Speaker is significantly more committed to the embedded *p* when *p* appears in the indicative than when *p* appears in the subjunctive.
 - For non-factive/**V2 verbs**, mood choice has **no significant effect** on speaker commitment to *p*.

5.3. Proposal for Polarity Mood

■ Our analysis has two components:

- The same **lexical entries for IND and SUBJ** that we used for Intensional Mood
- Since V1 verbs are **soft presupposition triggers** but V2 verbs are not, we need an analysis of soft presuppositions drawing a distinction between the two: Romoli (2015).

■ Romoli (2015):

- Romoli (2015) derives so-called “soft presuppositions” not as lexical presuppositions but as scalar conversational implicatures arising from the interaction of:
 - i) A set of ordered alternatives invoked by the soft presupposition trigger
 - ii) Exhaustivity operator EXH: (95)

(95) $\llbracket \text{EXH} \rrbracket (\text{Alt}(p))(p)(w) = p(w) \wedge \forall q \in \text{Alt}(p) [p \not\subseteq q \rightarrow \neg q(w)]$
 I.e., EXH asserts the pronounced alternative and negates all the stronger alternatives.

- **V1 verbs** like *know*, being soft presupposition triggers, are associated with a set of ordered alternatives:

(96) a. John knows that Mary runs.
 b. Alt: { KNOW(j, RUN(m))_{str}, RUN(m)_{wk} },
 where KNOW(j, RUN(m)) means ‘Mary runs and John knows it’.

- In negative environments, the entailment relation is reversed, giving (97b) below. Exhaustification applies to the sentence and we obtain (97c).

(97) a. John doesn’t know that Mary runs.
 b. Alt: { $\neg$ KNOW(j, RUN(m))_{wk}, $\neg$ RUN(m)_{str} }
 c. Exh: $\neg$ KNOW(j, RUN(m)) $\wedge$ RUN(m)

- Crucially, **V2 verbs** like *say*, not being soft presupposition triggers, do not give rise to ordered alternatives: (98). Since the embedded proposition *p* does not come into play as a stronger alternative in (99b), exhaustification applies vacuously and we only derive the original literal meaning (99c).

(98) { SAY(x,p) }

(99) a. John didn’t say that Mary runs.
 b. Alt: { $\neg$ SAY(j, RUN(m)) }
 c. Exh: $\neg$ SAY(j, RUN(m))

UPSHOT 5

V1 verbs	V2 verbs
John doesn’t know that Mary runs.	John didn’t say that Mary runs.
$\rightsquigarrow \neg \text{KNOW}(j, \text{RUN}(m)) \wedge \text{RUN}(m)$	$\rightsquigarrow \neg \text{SAY}(j, \text{RUN}(m))$

This derives a crucial difference between V1 vs V2.
 However, the effect of mood still needs to be factored in.

(100) a. $\llbracket \text{IND}_2^{\text{pro1}} \rrbracket^g$ is defined only if $\text{content}(g(1))$ is **SIMPLEX** and $g(2) \in \text{Dom}(\text{content}(g(1)))$;
 if defined, $\llbracket \text{Ind}_2^{\text{pro1}} \rrbracket^g = g(2)$
 b. $\llbracket \text{SUBJ}_2^{\text{pro1}} \rrbracket^g$ $g(2)$ (=39)

(101) a. John doesn't know.IND [that Mary runs.IND/SUBJ]
 b. Exh: $\neg\text{KNOW}(j, \text{RUN}(m)) \wedge \text{RUN}(m)$
 c. $\text{ASSERT}(\neg\text{KNOW}(j, \text{RUN}(m))) \wedge \text{OPERATOR}(\text{RUN}(m))$

first conjunct
second conjunct

(103) Proposition conveyed by the second conjunct in (90c) before mood is added:
 $\lambda_{w_0}.$ OPERATOR_{w₀} ($\lambda_{w'}:$...???... . RUN_{w'}(m))

- For the first conjunct, the intended domain of quantification is the one introduced by *know*, which is simplex and thus licenses IND: (104)
- For the second conjunct, the speaker uses the presuppositionally heavier IND to convey that the intended force operator is ASSERT (since ASSERT licenses IND, as in (105), and no other operator would, as in (106)). Thus, by using IND, the Speaker explicitly presents herself as committed to the embedded proposition *p*.

(105) Second conjunct with ASSERT as operator: ✓
 a. $\lambda w_0. \text{ASSERT}_{w_0}(\lambda w': \text{content}(g(1)) \text{ is SIMPLEX and } w' \in \text{Dom}(\text{content}(g(1))). \text{RUN}_{w'}(m))$
 b. $\lambda w_0. \forall w \in \cap \text{Rpg}(w_0): w \in [\lambda w': \text{Rpg}(w_0) \text{ is SIMPLEX and } w' \in \cap \text{Rpg}(w_0). \text{RUN}_{w'}(m)]$

21

■ If the embedded CP is in **SUBJ**:

- Neither for the first conjunct nor for the second does SUBJ introduce a presupposition: (107)-(108).
- By having SUBJ in the second conjunct the speaker presents herself neither as committed nor as uncommitted to p .

(107) First conjunct: ✓

$\lambda w_0. \text{ASSERT}_{w_0}$
 $(\neg \text{KNOW}(j, \lambda w': \text{content}(g(1)) \text{ is SIMPLEX and } w' \in \text{Dom}(\text{content}(g(1))). \text{RUN}_{w'}(m)))$

(108) Second conjunct with any operator OPR: ✓

a. $\lambda w_0. \text{OPR}_{w_0} (\lambda w': \text{content}(g(1)) \text{ is SIMPLEX and } w' \in \text{Dom}(\text{content}(g(1))). \text{RUN}_{w'}(m))$

■ In other words:

UPSHOT 5

V1 verbs	V2 verbs
John doesn't know that Mary runs.	John didn't say that Mary runs.
$\rightsquigarrow \neg \text{KNOW}(j, \text{RUN}(m)) \wedge \text{RUN}(m)$	$\rightsquigarrow \neg \text{SAY}(j, \text{RUN}(m))$

+

UPSHOT 3

$\text{ASSERT}_{\text{Rpg_x}(w_0)} [p. \text{IND} / p. * \text{SUBJ}]$	$\text{COMM/WISH/SURP}_{\langle f(w_0), f(w_0) \rangle} [p. * \text{IND} / p. \text{SUBJ}]$
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=

This derives the different strength in presupposition projection with V1 verbs:
 Indicative explicitly signals that the intended operator is ASSERT and, thus, expresses speaker commitment to embedded p .
 Subjunctive signals that the intended operator is not ASSERT, leaving open whether the speaker is committed to p or not.

■ We turn now to **V2 verbs**.

- (109) a. John didn't say.IND [that Mary runs.IND/SUBJ]
 b. Exh: $\neg \text{SAY}(j, \text{RUN}(m))$
 c. $\text{ASSERT}(\neg \text{SAY}(j, \text{RUN}(m)))$

first conjunct only!

■ If the embedded CP is in **IND**:

The embedded partial proposition suffices and thus IND is licensed.

- (110) First conjunct: ✓

- a. $\lambda w_0. \text{ASSERT}_{w_0}$
 $(\neg \text{SAY}(j, \lambda w': \text{content}(g(1)) \text{ is SIMPLEX and } w' \in \text{Dom}(\text{content}(g(1)))) . \text{RUN}_{w'}(m))$
 b. $\lambda w_0. \forall w \in \cap \text{Rpg}(w_0):$
 $\neg \forall w'' \in \cap \text{Rpg}_j(w): w'' \in [\lambda w': \text{Rpg}_j(w) \text{ is SIMPLEX and } w' \in \cap \text{Rpg}_j(w). \text{RUN}_{w'}(m)]$

■ If the embedded CP is in **SUBJ**:

The embedded total proposition imposes no condition. Assuming that the effects of Maximize Presupposition can be can be suspended here, SUBJ is licensed.

- (111) First conjunct: ✓

- $\lambda w_0. \text{ASSERT}_{w_0}$
 $(\neg \text{SAY}(j, \lambda w': \text{content}(g(1)) \text{ is SIMPLEX and } w' \in \text{Dom}(\text{content}(g(1)))) . \text{RUN}_{w'}(m))$

■ In other words:

UPSHOT 5

V1 verbs	V2 verbs
John doesn't know that Mary runs. $\leadsto \neg \text{KNOW}(j, \text{RUN}(m)) \wedge \text{RUN}(m)$	John didn't say that Mary runs. $\leadsto \neg \text{SAY}(j, \text{RUN}(m))$

+

UPSHOT 3

$\text{ASSERT}_{\text{Rpg}_x(w_0)} [p. \text{IND} / p. * \text{SUBJ}]$	$\text{COMM/WISH/SURP}_{\langle f(w_0), f(w_0) \rangle} [p. * \text{IND} / p. \text{SUBJ}]$
--	---

=

This derives the lack of significant impact of mood choice on speaker commitment to p with V2 verbs.

6. OUTLOOK

■ Combining the following ingredients from different analyses in the literature, we have proposed the lexical entries in (112) for mood morphology in Spanish:

- Ingredient ①: The presuppositional constraint is formulated in terms of **simplex content of an attitude situation s** (as in Approach A), not in term of the local context set (as in Approach B).
- Ingredient ②: Mood introduces a **world pronoun** (as in Approach B), not quantification over worlds (as in Approach A).
- Ingredient ③: **Subjunctive** mood is the presuppositionally **unmarked** form (as in Approach B), not a presuppositionally contentful form (as in Approach B).

$$\begin{array}{ll}
 (112) \text{ a. } \llbracket \text{IND}_2^{\text{prol}} \rrbracket^g & \text{is defined only if } \text{content}(g(1)) \text{ is SIMPLEX and} \\
 & \text{if defined, } \llbracket \text{Ind}_2^{\text{prol}} \rrbracket^g = g(2) \\
 \text{b. } \llbracket \text{SUBJ}_2^{\text{prol}} \rrbracket^g & g(2) \hspace{10em} (=39)
 \end{array}$$

■ The meaning contribution of mood is kept uniform across constructions in the grammar, including:

- Intensional mood selected by attitude verbs
 $\Leftrightarrow$ Crucial for choice of ingredient ①: ‘hope’, ‘be possible/probable’
- Mood alternation in Relative Clauses
 $\Leftrightarrow$ Crucial for choice of ingredient ②: de re and de dicto readings but no third reading
- Mood alternation in Conditional Clauses
 $\Leftrightarrow$ Crucial for choice of ingredient ③: suppletion; non-counterfactual interpretation of subjunctive-marked conditionals in certain pragmatic contexts

■ Steps have been taken towards attaining a better understanding of polarity mood:

- Experimental study:
V1 verbs (cognitive factives) but not V2 verbs (cognitive non-factives) show a significant effect of mood choice: IND leads to a significantly higher speaker commitment to p than SUBJ in the case of V1 verbs but not in the case of V2 verbs.
- Proposed analysis:
Lexical entries in (112) + Romoli (2015) on soft presupposition triggers

■ Some important questions on polarity mood are left for future research, including:

- In the case of V2 verbs, if speaker commitment does not correlate with mood choice, what interpretive distinction correlates with it (if any)?

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